

Quantification of sovereign risk: Using the information in equity market prices

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Received 12 June 2005; received in revised form 2 September 2005; accepted 9 September 2005
Available online 24 October 2005

Abstract

This paper proposes a new approach to quantify the sovereign risk. We use the information of the stock price index as a proxy for the equity value of the country, and introduce a size parameter, which is a conversion factor of the stock price index to the equity value of the country, and adopted the extended Black–Scholes–Merton option-pricing model for the calculation of the probability of default. Our model is applied on two countries, Argentina as the case of debt crisis and Thailand as currency crisis. We demonstrated a capability of our model as an early warning indicator for both crises.

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JEL classification: C53; F34

Keywords: Sovereign risk; Probability of default; Stock price index; Option-pricing model

1. Introduction

It was commonly believed that the sovereign will never fall into default, however, the debt defaults in Latin countries happened in the early 1980s have reminded us an importance of the risk management even for sovereign countries. A number of models have been proposed for the evaluation of the sovereign risk. Most of these models are based on a statistical approach, for example, the multivariate discriminant analysis proposed firstly by Frank and Cline (1971) and the logistic regression analysis (e.g., Feder and Just, 1977). These models require various macroeconomic data as explanatory variables, however, it is inevitable that they cause a non-

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trifling time delay until the updated data become available. In addition, these data will be updated no more than once a year. These facts cause serious difficulties for the evaluation of the sovereign risks in timely manner.

Redeeming these shortcomings in the statistical approaches, the elaborate analysis by professional researchers seems to be helpful. In fact, the sovereign ratings issued by the rating agencies, such as Moody's or Standard and Poor's (S&P), are quite common as the reference of the sovereign risk. They conduct exclusive interviews to both politicians and officers in the country, and make macroeconomic forecast based on confidential information. Therefore, it is necessarily considered that the sovereign ratings should be a better reference than the simple quantitative statistical approach. As a number of papers pointed out,¹ however, the rating agencies indeed failed to predict the Asian crises that happened in 1997.

In the field of the corporate credit risk measurement, the Black–Scholes–Merton (BSM) option-pricing model² is fruitfully accepted, which is originally proposed by [Black and Scholes \(1973\)](#) and [Merton \(1974\)](#). Some of papers have applied the BSM option-pricing model to the measurement of sovereign risk. [Gennotte et al. \(1987\)](#) and [Clark \(1991\)](#) estimated the probability of default by employing an accumulated fixed capital formulation as the underlying asset in the option-pricing model. [Kirmann and Maltritz \(2003\)](#) showed the foreign exchange reserves and discounted that future net export incomes are the good proxy for the underlying asset or the ability to pay.

[Chesney and Morisset \(1992\)](#), [Claessens and Pennacchi \(1991\)](#) and [Cumby and Evans \(1997\)](#) estimated the probability of default based on a stochastic model. Meanwhile, [Bhanot \(1998\)](#), [Izavorski \(1998\)](#), [Copeland and Jones \(2001\)](#) and [Merrick \(2001\)](#) proposed a method to estimate the probability of default from the pricing model of sovereign bonds. [Lehrbass \(2000\)](#) proposed a pricing model of Eurobonds by use of equity indices.

In this paper, we propose a new model to quantify the sovereign risk. Our model is constructed based on the BSM option-pricing model, which enables us to evaluate sovereign risk in timely manner. We use a stock price index, as a proxy for the equity value of the country, and a sovereign bond-price.

We treat both debt default (debt crisis) and currency crisis. Although most of previous studies dealt with these two crises as separate issue, as [Reinhart \(2002\)](#) reported, there is a close relationship between defaults and currency crises, and indeed [Reinhart \(2002\)](#) showed the 58% samples of debt crises have fallen into defaults within twenty-four months after the currency crisis. We will examine the effectiveness of our model for predicting both debt and currency crises.

The remainder of this paper is organized as follows. Section 2 describes the usefulness of the information from the stock market in evaluating the credit risk, and its potential application to the sovereign risk. Section 3 describes the details of our model. Some results and reviews are summarized in Section 4. Our discussion and future extensions are in Section 5.

2. Proxy for the equity value of country

2.1. The concept of BSM model for risk analysis

[Fig. 1](#) shows the payoff function to the equity holder and debts holders. Let D_1 and D_2 be the face values of senior and junior debt, respectively, and D be the total liability, i.e., $D = D_1 + D_2$. E

¹ For example, see [Goldstein et al. \(2000\)](#), [Kraussl \(2000\)](#).

² This approach is sometimes referred to as “structural approach.”

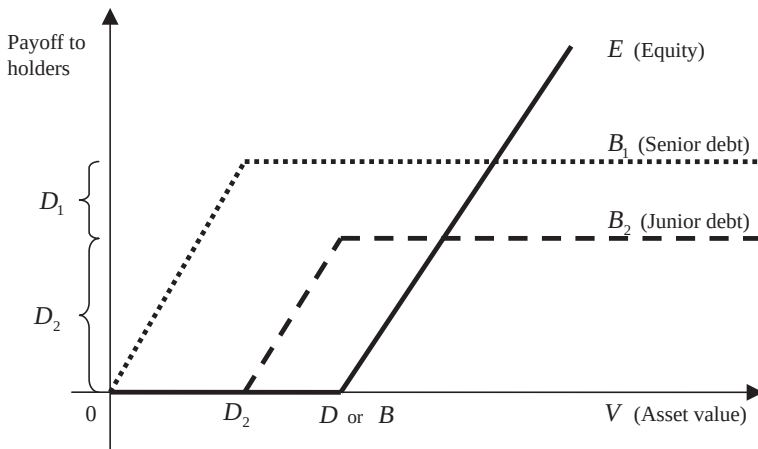


Fig. 1. Conceptual drawing of the payoff function to the equity holder and debts holders. D_1 and D_2 is the face values of senior and junior debt, respectively, and D is the total liability, $D = D_1 + D_2$. E denotes the payoff to the equity holder, and B_1 and B_2 stand for the payoff to the senior and the junior debt holder, respectively. $B = B_1 + B_2$ is the total payoff to the debt holders.

denotes the payoff to the equity holder, and B_1 and B_2 stands for the payoff to the senior and the junior debt holder, respectively. Then $B \equiv B_1 + B_2$ is the total payoff to the debt holders, which can be also recognized as the market value of the total debt. As shown in the Fig. 1, the market value of the debt, B is the put option on the value of the asset V , which is given by $B = \min(V, D)$. Taking the seniority of the debts into consideration, the values of the senior and junior debt will be $B_1 = \min(V, D_1)$ and $B_2 = \min(V - B_1, D_2)$, respectively, since the junior debt D_2 is subordinated to the senior debt D_1 , their payoff or market value B_2 is the residue of the asset value after the payoff to the senior debt holder. The market value of the equity, S is subordinated to the senior and junior debts, thus, it will be the residue of the value of the asset after the payoff to the all debt holders, $E = \max(0, V - B)$ or equivalently $E = \max(0, V - D)$.³ Therefore, the market value of asset V can be calculated from the total liability D and the market value of equity E which can be observed in the stock market.

Under the BSM option-pricing model, the probability of default is given as the likelihood that the market value of the asset becomes less than the market value of its total liability at the maturity T , i.e., $\Pr(V_T \leq D)$. This will be discussed in detail in Section 3.1.

2.2. Market value of equity in country

In applying the BSM option-pricing model to sovereign risk, either the market value of the asset or the market value of equity in the country is necessary, unfortunately, both of them are not readily available. Instead, we can obtain the information from stock market, which may be useful as a proxy for the equity value of the country.⁴ The equity market is not established in all countries, however, it covers over 70 countries including all financially major countries. We examine the relationship between the stock price index and the equity value of the country in following.

³ This payoff function shows the equity value is a call option on the value of the asset with a strike price equal to the total liability.

⁴ Pettis (2001) discussed the country's "underlying asset value," and stated, "one possibility is that some proxy for the country's equity markets – such as a broad-based equity index – could also serve as a proxy for the underlying assets."

Table 1
Macroeconomic balance sheet at time T

Asset	Liability
☐ Total reserves. ☐ Sum of gross fixed investment from 0 to the end of period T . ☐ Sum of increase in stocks from 0 to the end of period ☐ Less sum of depreciation from 0 to the end of T .	☐ Net short-term foreign liabilities outstanding at the end of T . ☐ Net medium and long-term foreign liabilities outstanding at the end of T . ☐ Net portfolio and direct investment outstanding at the end of T .
	Equity
	☐ Sum of economies retained earnings from 0 to the end of period T .

2.3. Stock price index and the equity value of country

There exists some methods to analyze the equity values of countries. United Nations proposed Systems of National Accounts (abbreviated to SNA), which is constituted of 5 major accounts such as National Income Accounts, Balance of Payments and National Balance Sheets. The National Balance sheets contains the information of equity value of the country, however, it is provided by a limited number of countries.

Clark (1991) proposed the concept of macroeconomic balance sheet. The macroeconomic balance sheet is prepared by the time series data of National Income Accounts and Balance of Payments, which are available for most of countries. The macroeconomics balance sheet is constructed from three parts, the asset, the liability and the equity, as shown in Table 1. The equity value of the country can be calculated as “assets” minus “liabilities.”

We calculated the macroeconomic balance sheets for Argentina and Thailand, where the depreciation rate for the fixed asset is assumed to be 10% each year, and the accumulations are made retroactively from 1970 to 1999, all data are denominated in US dollars. The amount of liability is collected from Global Development Finance (GDF) issued by World Bank. Fig. 2 shows the transitions of the equity values of the country obtained from above calculation and the stock price index for Thailand and Argentina. The stock price indices are obtained from Morgan Stanley Capital International’s (MSCI) capital indices as an average through the year. Table 2 shows the Pearson correlation coefficients between stock price index and the equity value of the country, where we demonstrate the same calculation for 10 countries in total.

They show that the stock price index is tightly related to the equity value of the country. In consideration of this concordance, this will be good evidence that the stock price index will be a good proxy of the equity value of the country.

2.4. Transformation of stock price index into the equity value of the country

Our model employs the stock price index of the country. The stock price index is a relative value measuring from a specific origin date, and depends on the market. In the case of MSCI’s stock index, for most of countries, 31st of December, 1987, is the origin of the index as 100. Given a

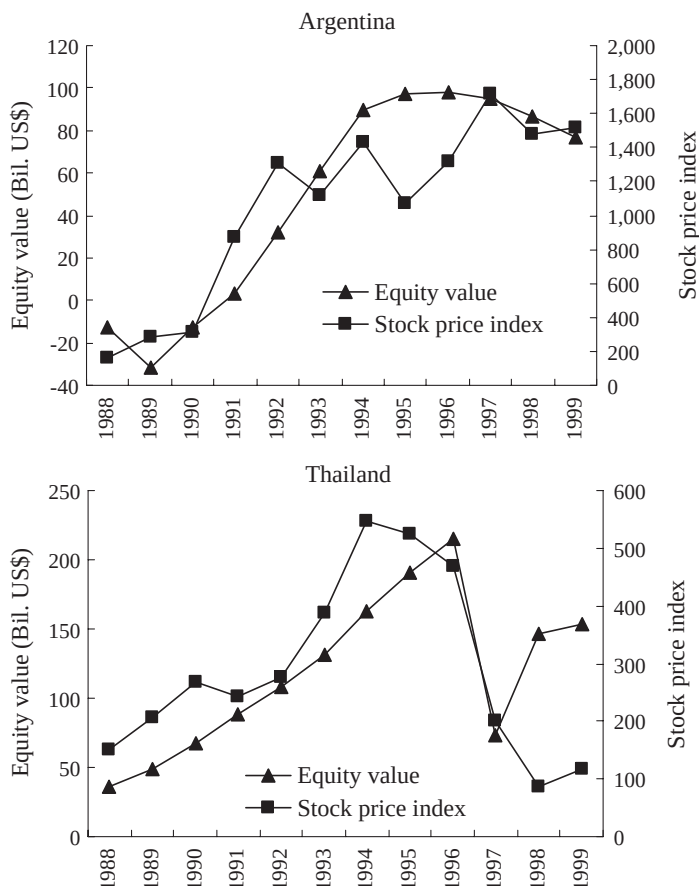


Fig. 2. The transitions of stock price index and equity value for Argentina and Thailand. Stock price index is obtained from MSCI's equity indices as an average through the year. Equity values are calculated based on the concept of macroeconomic balance sheet proposed by Clark (1991).

country, in order to transform the stock price index into the equity value E_t of the country at time t , which is an absolute value, we employ the following equality:

$$E_t = \alpha S_t \quad (1)$$

where S_t is the stock price index, and α stands for a constant parameter for the country. α can be understood as a specific scaling parameter between E_t and S_t . Each country has its own scaling parameter α and we call α as “size parameter”, and assume that α is unique to the country and constant in time.

3. Evaluation of sovereign risk

3.1. Option-pricing model

In the BSM option-pricing model, the asset value V_t of a country at time t is assumed to follow the stochastic differential process:

$$dV_t/V_t = \mu_V + \sigma_V dW_t$$

Table 2

Pearson correlation coefficients between the stock price index and the equity value of the country

Country	Correlations
	1988–1999
Thailand	64.4%
Indonesia	61.9%
Malaysia	65.3%
Philippines	80.2%
Argentina	87.0%
Brazil	87.5%
Mexico	77.3%
Chile	83.2%
Pakistan	72.7%
	(1993–1999)
Peru	72.3%
	(1993–1999)

The stock price index is obtained from MSCI's equity index. The index is available from the end of 1987, except Pakistan and Peru, which is available after 1993.

where μ_v is a growth rate of the asset, σ_v is a volatility of the asset, and W_t is a Winner process. In applying Ito's formula, the natural log of asset value at time T is distributed normally as follows,

$$\ln V_T \sim \text{Norm} \left[\ln V_t + \left(\mu_v - \frac{\sigma_v^2}{2} \right) \tau, \sigma_v^2 \tau \right], \quad (2)$$

where $\text{Norm}[X, Y]$ is the normal distribution function with mean X and variance Y , and $\tau \equiv T - t$ is the risk horizon. We assume that the country will fall into default if the asset value V_T becomes less than the value of the foreign liability D_T at the maturity date T . The expected probability of default $P_{t(T)}$ will be given in the following form:

$$\begin{aligned} P_{t(T)} &= \Pr(V_T \leq D_T | V_t) = \Pr(\ln V_T \leq \ln D_T | \ln V_t) \\ &= N \left(\frac{\ln D_T - [\ln V_t + (\mu_v - \sigma_v^2/2)\tau]}{\sigma_v \sqrt{\tau}} \right) = N(-d_2) \end{aligned} \quad (3)$$

where

$$d_2 \equiv \frac{\ln(V_t/D_T) + \left(\mu_v - \frac{\sigma_v^2}{2} \right) \tau}{\sigma_v \sqrt{\tau}}$$

and $N(\cdot)$ is the cumulative normal distribution function. The market value of the equity E_t at time t becomes:

$$E_t = V_t N(d_1) - D_T e^{-\mu_v \tau} N(d_2). \quad (4)$$

where $d_1 \equiv d_2 + \sigma_v \sqrt{\tau}$.

Eqs. (3) and (4) contain three unknown parameters μ_v , σ_v and V_t . We employ an iterative method proposed by [Chesney and Morisset \(1992\)](#) for estimating parameters. In our model, we firstly set a size parameter α some value, and solve the Eqs. (3) and (4). Then, the size parameter will be calibrated properly by use of the bond-prices.

3.2. Theoretical bond-price

For time t , let $\bar{p}_{t(s)}$ be the probability that the bond first falls into default at time $t+s$ under the condition that there is no default occurred prior to $t+s$. We hereby assume that the conditional probability of default $\bar{p}_{t(s)}$ has a flat term structure, $\bar{p}_{t(s)} = p_t$ for all s .⁵ The probability of default from t to $t+s$ would be $P_{t(s)} = 1 - (1 - p_t)^s$. The present value of the fixed-rate coupon bond at time t with NC periods to maturity will be given by the following formula:

$$B_t = \sum_{i=1}^{NC} \{ (1 - P_{t(i)}) d_{t(i)} cF \} + (1 - P_{t(NC)}) d_{t(NC)} F + \sum_{i=1}^{NC} (\beta_t p_t d_{t(i)} F) \quad (5)$$

where $d_{t(k)}$ is the discount factor and given in $d_{t(k)} = 1 / (1 + r_{t(k)})^k$, $r_{t(k)}$ is the yield rate of a U.S. Treasury zero-coupon bond maturing at $t+k$,⁶ c is a coupon rate, F is the face value of the bond, β_t is the recovery rate. We assume that the recovery rate β_t is constant through to the maturity but varying on time t , and the recovery value $\beta_t F$ will be paid immediately upon its default. The first term of the RHS of Eq. (5) means the cash flow from coupon payments, the second term is the payment of principle, and the last term is the recovery amount when it defaults.

3.3. The liabilities and their maturities

The data of foreign liabilities are collected from GDF. However, we need to pay attention to these data when applying to Eq. (3), because the foreign liability of the country is a mixture of various face values and maturities. In accordance with Clark and Marois (1996), we consider that the sum of all the contractual debt service payments coming due in the future including interest and principal is a good estimation of D_T , and the time to maturity, τ is estimated as the Macaulay duration⁷ of such payments.

3.4. Optimization of size parameter

In order to determine the size parameter α and recovery rate β_t , we employed the following algorithm.

- Step 1 Set a size parameter $\tilde{\alpha}$ appropriately, and calculate the probability of default $P_{t(T)}$ by the Eqs. (3) and (4).
- Step 2 The theoretical bond-price $\tilde{B}_t$ at time t is obtained from Eq. (5) as the function of $\tilde{\alpha}$ and an assumed recovery rate $\tilde{\beta}_t$, i.e., $\tilde{B}_t = f(\tilde{\alpha}, \tilde{\beta}_t)$. The optimal recovery rate β_t^* will be given as the point of minimization between the estimated theoretical bond-price and the observed bond-price in the market B_t^M . The grid search method is used to obtain theoptimal recovery rate.⁸

$$\beta_t^* = \arg \min_{\tilde{\beta}_t} \left\{ (B_t^M - f(\tilde{\alpha}, \tilde{\beta}_t))^2 | \tilde{\alpha} \right\}. \quad (6)$$

⁵ Cumby and Evans (1997) and Merrick (2001) reported that the probability of default has a term structure. However, the form of the structure cannot be described explicitly and heavily indebted to the data for parametric estimations, which will cause difficulty to obtain relevant results. For simplicity purpose, we assumed the flat term structure in our model.

⁶ The yield rate of a U.S. Treasury zero-coupon bond is obtained by linear interpolation of spot rates with three and six months and one, two, five, ten and thirty years.

⁷ Macaulay duration is calculated as the time-weighted discount cash flow.

⁸ Considering heavy computational burden, we searched the recovery rate from 0 to 1 by every 5% step.

Step 3 The parameter calibration process will be repeated for the certain periods ($0 \leq t \leq T_c$), and we hereinafter call it the calibration periods. The optimal size parameter α^* will be given as the point of minimization for the sum of the residual values between the theoretical bond-prices $\tilde{B}_t^* = f(\tilde{\alpha}, \beta_t^*)$ and the observed bond-prices in the market B_t^M under the optimal recovery rate β_t^* ,

$$\alpha^* = \arg \min_{\tilde{\alpha}} \left\{ \sum_{t=0}^{T_c} (B_t^M - f(\tilde{\alpha}, \beta_t^*))^2 | \beta_t^* \right\}. \quad (7)$$

Since α^* cannot be determined analytically, the grid search method is used to obtain the optimal value. The optimal value of the size parameter α^* is the country specific and constant in time.

4. Results

4.1. Selected countries

We apply the model on two countries, Argentina and Thailand. Argentina fell into default on their foreign debt in November 2001, and Thailand faced the sharp devaluation of their currency in July 1997. Both cases left the serious impacts on world equity and currency markets. These will be good cases to test whether our model has a capability as an early warning indicator for both debt and currency crisis.

4.1.1. Stock price indices

Several stock price indices exist for these countries. In general, the stock exchange market in each country provides their own index, and in addition, some security companies issue their unique indices based on their own methodology. For example in the market of United States, there are S&P 500, DJIA, NASDAQ composite and more indices. Each index has different process to select companies composing the indices. In the emerging markets, there are two major indices, Morgan Stanley Capital International (MSCI) and S&P (formerly International Finance Corporation). We use MSCI's equity index denominated in US dollar. The indices are announced on daily basis, and cover 23 developed and 27 emerging countries. MSCI's equity index captures 60% of the total market capitalization of the country.

4.1.2. Bond-prices

Table 3 shows the details of sovereign bonds used for the calibration. All data are collected from DataStream terminal. Each country usually issues a number of bonds with different

Table 3
Details of the sovereign bonds used for the parameter calibrations

	Argentina	Thailand
Issuer	Government of Argentina	Government of Thailand
Coupon	9.375%	7.750%
Maturity	Dec. 20, 2003	Apr. 15, 2007
Currency	U.S. Dollar	U.S. Dollar
Data availability	Jan. 96–Dec. 00	Apr. 97–Dec. 00

These data are collected from DataStream terminal.

Table 4

Size parameters and search steps

	Optimal size parameter α^*	Search step for size parameter α	Search step for recovery rate β
Thailand	80	5	5%
Argentina	7	1	5%

coupons and maturities. The reasons we select these bonds are as follows; first, these bonds are ordinal fixed coupon bonds. Floating rate bond or collateralized bond like Brady bond is not suitable for the calculation of Eq. (5).⁹ Second, the currency is denominated in US dollar. Finally, these bond-prices are available for longer periods than other sovereign bonds that we can obtain.

4.2. Default probabilities

Table 4 shows the optimal size parameters α^* calculated from Eq. (7). The search step of the size parameter is a crucial for the accuracy of the results, but it is also proportional to the calculation burden. We have determined an appropriate search step country by country, i.e., 5 for Thailand and 1 for Argentina.

Fig. 3 shows the probability of default calculated from Eq. (3) based on the optimal size parameter α^* for the period from Jan. 94 to Dec. 2001, and the stock price index is also shown as a reference. The perpendicular bold dashed line shows the line of the debt default for Argentina and the currency crisis for Thailand.

Argentina has defaulted on their foreign debt in November 2001. The probability of default keeps high around 10% to 15% since the early 1995, and it sharply climbs out from July 2001, which was 4 months prior to the debt default. At the debt default in November 2001, the probability of default pointed to the peak of 28%. Meanwhile, for Thailand, the default probabilities start to increase from early 1996. It was almost one year prior to the crisis in July 1997, and it sharply climbs out after the crisis, and became more than 30% at the peak in 1998. After the end of 1999, when the currency crisis seemed to be settled, the probability of default becomes stable around 5% to 10%.

4.3. Recovery rate

Fig. 4 shows the recovery rate optimally estimated from Eq. (6). This may be called as the implied recovery rate.¹⁰ The fundamental meaning of the recovery rate in Eq. (5) is the fraction of the expected recovery amount and the face value of the bond when it falls into default. In our model, it also works as an adjustment factor for the price difference between estimated theoretical bond-price, $\tilde{B}_t$ and observed bond-price, B_t^M . In other words, the recovery rate will be endogenously determined in order to equalize the price of $\tilde{B}_t$ with B_t^M under the given default probability.

Fig. 5 compares the observed bond-price in the market with the theoretical bond-price in adopting the optimally estimated recovery rate, β_t^* . They indicate good agreements throughout the period. These results show that our bond-pricing model works well throughout the period.

⁹ Cumby and Pastine (2001) investigated the pricing of Brady bonds.

¹⁰ We cannot observe the recovery rate directly in the market data, but it is “implied” and must be estimated from the data.

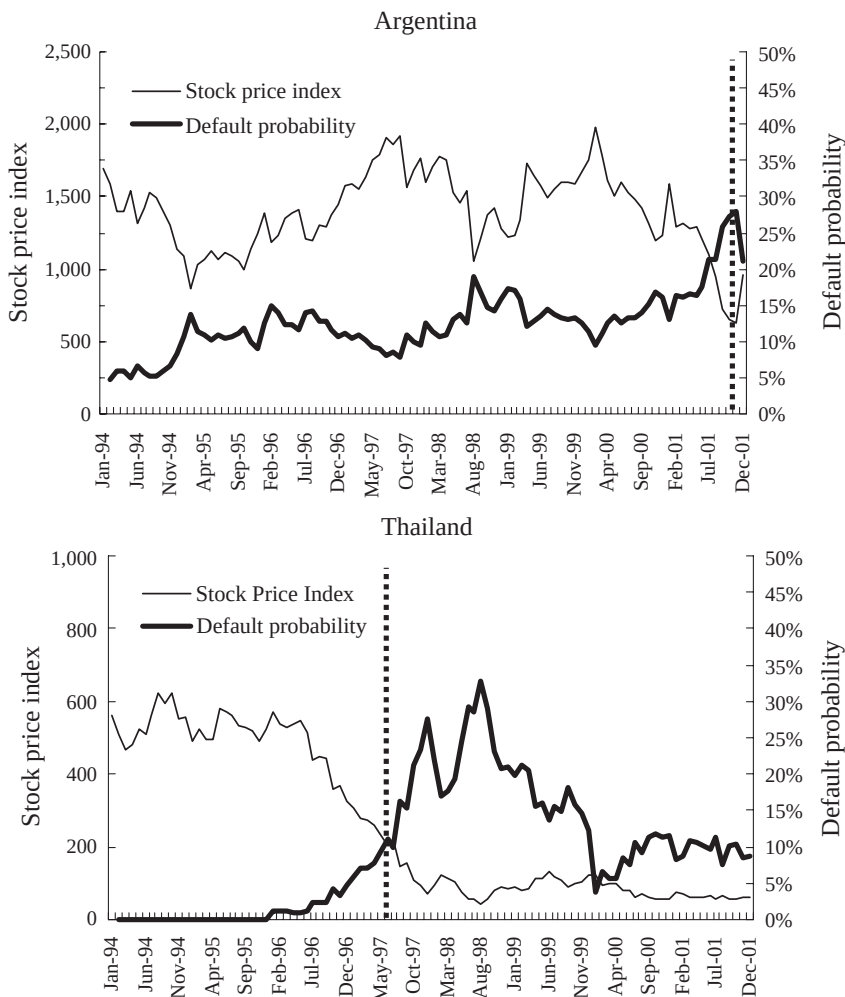


Fig. 3. Default probability and stock price index. The perpendicular bold dashed line shows the line of debt default for Argentina and the currency crisis for Thailand. The default probabilities are calculated from Eq. (3).

From scrutiny of Fig. 4, we can perceive that every country has a mean level of the recovery rate, for instance, the mean level of Argentina is around 60% to 70%, and Thailand is around 80% to 90%. As an examination, we fixed the recovery rates at 85% for Thailand and 65% for Argentina, the results are shown in Fig. 6. We will notice that the difference between theoretical and observed bond-prices shows an acceptable agreement throughout the period, however, the differences occasionally become wider, especially during the crisis, Asian currency crisis, Argentina debt crisis and interestingly Russian debt crisis in August 1998 as well.¹¹ These price differences vanished in a relatively short period. This means the bond-price observed in the market is very sensitive against the credit events happened even in

¹¹ Russia declared the payment rescheduling for their domestic debt, GKO(Gosudarstvennye Kratkosrochnye Obyazatel'stva).

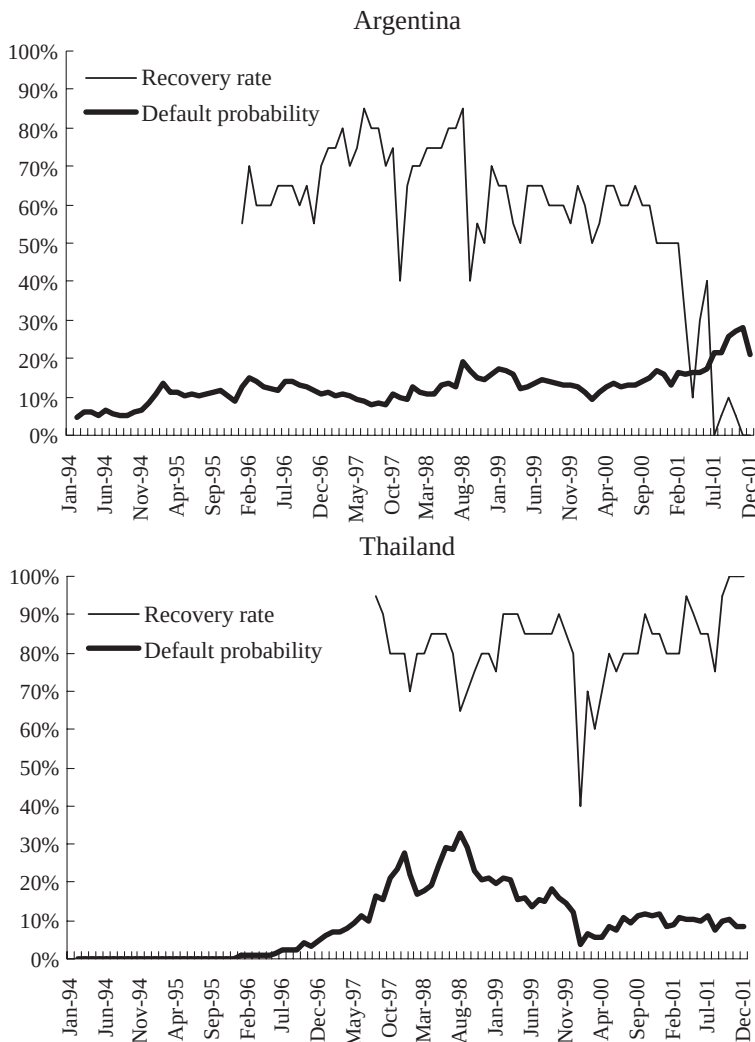


Fig. 4. Default probability and recovery rate. The recovery rates are optimally estimated from Eq. (6). Every country seems to have a mean level of the recovery rate, around 60% to 70% for Argentina, and around 80% to 90% for Thailand.

foreign countries. It responds to the credit events, and overshoots more than the price theoretically estimated from the probability of default.¹² However, the observed price reverts quickly to the level of theoretical price once the credit event has calmed. One of the reasons for this overshoot may be the liquidity problem of the bond during the crisis. In addition, we can point out another reason, pricing error of the market, which means that the investors may recognize the probability of default worse than that of theoretically expected. By analyzing this error, there may be a good opportunity to take an arbitrage trading. Another empirical study will be required for this area.

¹² We may say the situation as “deeply discounted.”

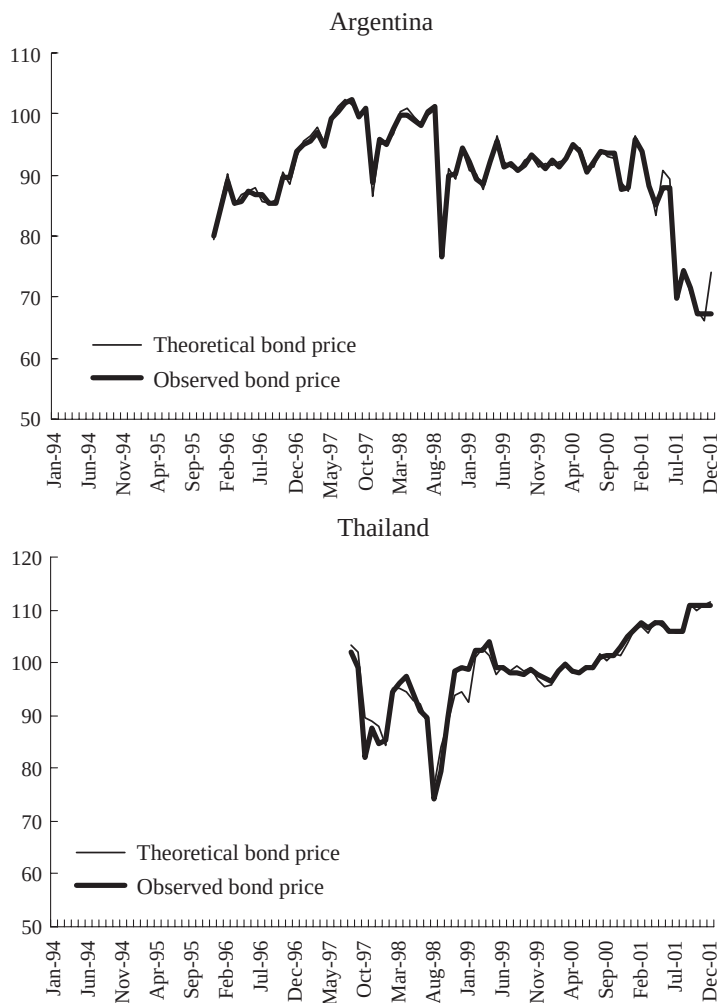


Fig. 5. Comparison between the observed bond-price in the market and the theoretical bond-price. The theoretical bond-price is calculated under the optimally estimated recovery rate.

4.4. Comparison with sovereign ratings

In Fig. 7, we compare the probability of default obtained by our model with S&P's sovereign ratings. In Argentina, our model keeps higher probability of default throughout the periods, but S&P's sovereign rating shows BB– and even upgraded to BB in early 1997. Then, S&P's sovereign rating started to downgrade from the end of 2000. On the other hand, the probability of default in our model started to climb out at the same time, which is almost one year prior to the debt default. Meanwhile, in Thailand, the S&P's sovereign rating is very high until the currency crisis, and then substantially downgraded after the crisis. On the other hand, our model shows that the probability of default started to climb out sharply from 1996 which is more than one year prior to the crisis. These evidences clearly show that our model has a better performance than the sovereign ratings as an early warning indicator.

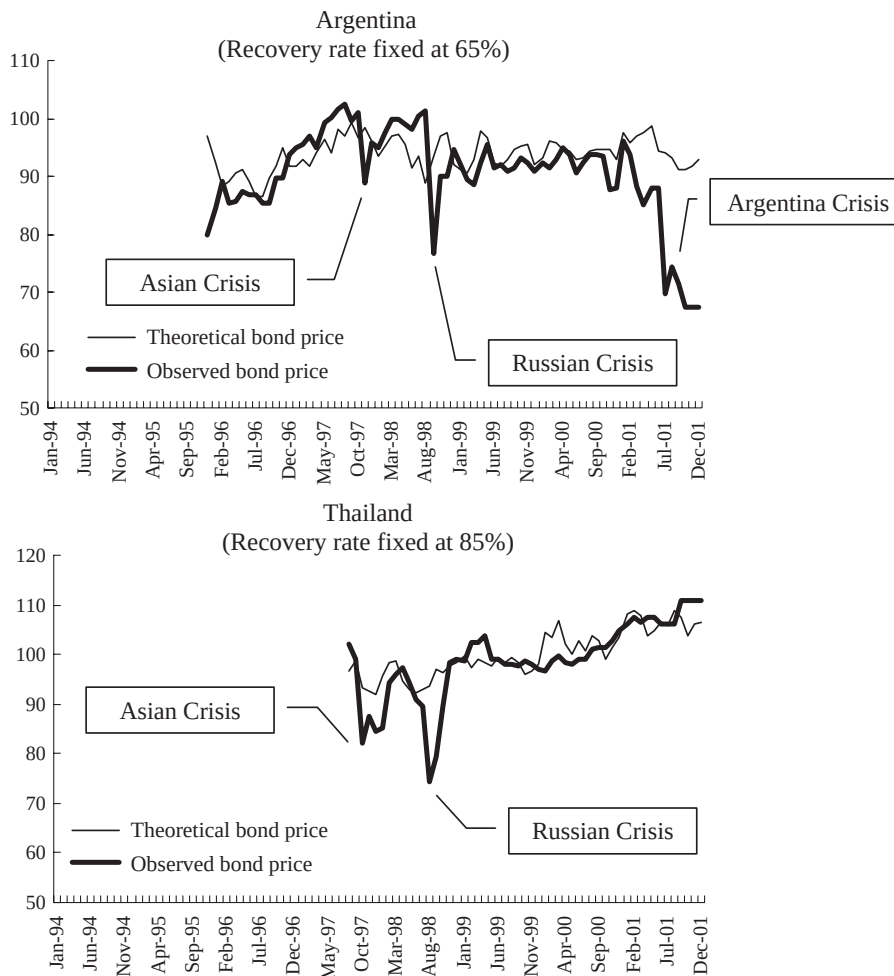


Fig. 6. Comparison between the observed bond-price in the market and the theoretical bond-price. The recovery rate is fixed at 65% for Argentina and 85% for Thailand. The large price gaps are observed when the crises have happened even in other countries.

4.5. Comparison with bond spreads

Fig. 8 shows the comparison of the probability of default obtained by our model with bond spread. The bond spread is calculated under following simplified formula:

$$\ln(B_t^{\text{NR}}/B_t^{\text{M}})/\tau \quad (8)$$

where B_t^{M} is the observed bond-price, and B_t^{NR} is the theoretical bond-price with no default risk under the assumption that the both bonds have the same cash flows and the time to maturity τ .

In both cases, the bond spreads show spikes to the credit events in Russian. In the case of Argentina, the bond spreads expand from 1.56% to 7.45% in one month, and 1.86% to 4.20% in Thailand. This is a peculiarity of bond spread and it is not a favorable nature from the viewpoint of risk evaluation. The desired nature is that the risk index starts to increase well in advance of

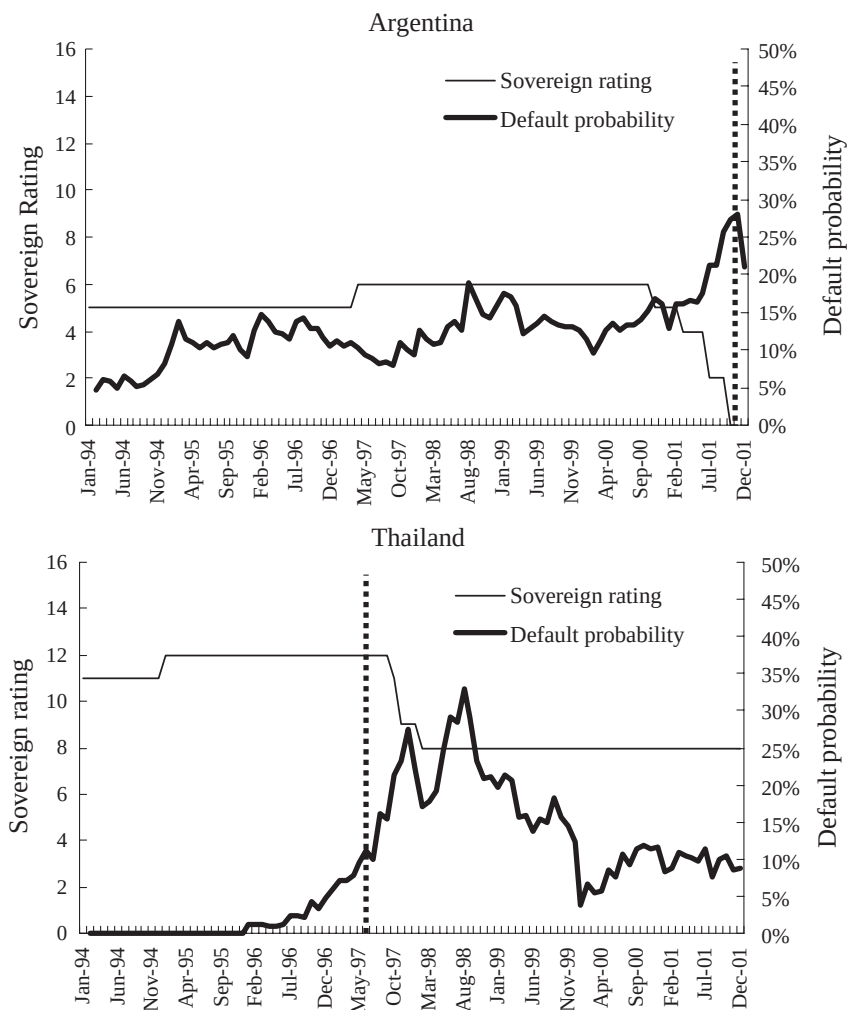


Fig. 7. Comparison between the default probability and sovereign rating. The sovereign ratings are based on S and P. The numbers are corresponded as follows: AAA=17, AA+=16, AA=15, AA-=14, ..., BB=6, BB-=5, B+=4, B=3, B-=2, CCC+=1, CCC and under=0. The perpendicular bold dashed line shows the line of the debt default for Argentina and the currency crisis for Thailand.

the events, and does not fluctuate too much during the calm period. In this sense, the results from our model have better nature as the risk index.

5. Discussion and future extensions

5.1. Parameter estimation

The bond-price issued by sovereign is used in our model. Each sovereign issues a number of bonds. If the parameters are calibrated only from a specific bond-price, the results may be biased. In order to avoid these unfavorable results, it is recommended to use a set of bonds.

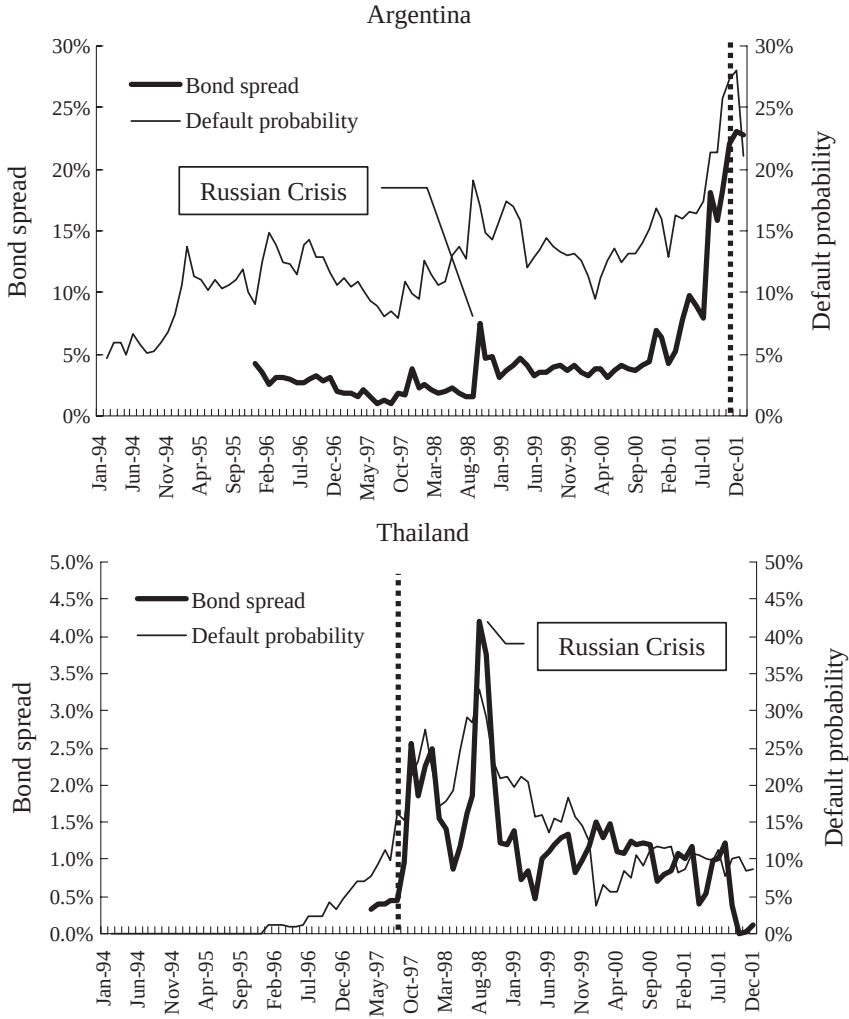


Fig. 8. Comparison between the default probability and bond spread. The bond spreads are calculated from Eq. (8).

In this case, the Eq. (7) could be modified to minimize the sum of price differences of a set of bonds, i :

$$\alpha^* = \arg \min_{\tilde{\alpha}} \sum_i \left\{ \sum_{t=0}^{T_c} (B_t^{M,i} - \tilde{B}_t^{*i})^2 | \beta_t^{*i} \right\} \quad (9)$$

where $\tilde{B}_t^{*i}$ is the function of $\tilde{\alpha}$ and β_t^{*i} . β_t^{*i} may be different bond by bond since each bond has his own features and attributes.

Furthermore, as we have seen in Section 4.5, the bond-price might be spiked or be overshot in the case that credit events have happened. This would cause unfavorable bias on the results if we make calibration only in short period. Therefore, it will be suitable to take the calibration period as long as possible.

5.2. Modeling default process

In Eq. (3), we have assumed that the default will occur only at the maturity date if the asset value is less than the value of foreign liability at the date. In consideration of practical situation, the default may occur even before the maturity date. In option pricing model, the former model is called as European type which is expressed as $P_{i(T)} = \Pr(V_T \leq D_T | V_T)$, and the later model is called as the first passage time model or knockout option model which is expressed as

$$P_{i(T)} = \Pr\left(\min_{t \leq u \leq T} V_u \leq D_T | V_t\right). \quad (10)$$

Ando and Marumo (2001) discussed that the probability of default by knockout option model generally indicates higher probability than that of European type model. From this result, we can expect that the optimal size parameter α^* will take a bigger number in the case that the knockout option model is employed instead of European type model. Since the bigger size parameter will decrease the probability of default, this will offset the increment of the probability of default by use of the knockout option model. From above discussion, it is worthwhile to note that the size parameter may vary depending on the adopted model.

5.3. Modeling of the willingness-to-pay

The sovereign may suspend their contractual obligations even if they are solvent, because the lenders have only limited power to punish sovereign borrowers and the international legal framework is not well developed for the recovery rules and processes from the defaulted borrowers. This will be a major factor to make distinct the sovereign borrower from corporate borrower. This was firstly discussed by Eaton and Gersovitz (1981), and is well known as the willingness-to-pay problem.

If there is willingness-to-pay problem, the payoff function to the sovereign bondholder will shift to the right, which is similar to the junior debt against the senior debt as shown in Fig. 1. Therefore, if we interpret the amount of D_1 as the level of unwillingness, we can discuss the willingness-to-pay problem under the same framework.

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